

Simplify Boolean Functions - 1

Unit – 2: Digital Electronics

**IN THE FOLLOWING TRUTH TABLE X AND Y ARE INPUTS
AND B AND D ARE OUTPUTS:**

Input		Output	
x	y	B	D
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

ANSWER THE FOLLOWING QUESTIONS:

(I) WRITE THE SOP EXPRESSION FOR D.

(II) WRITE THE POS EXPRESSION FOR B.

(I) WRITE THE SOP EXPRESSION FOR D.

x	y	D	Min Terms
0	0	0	
0	1	1	$x'y$
1	0	1	xy'
1	1	0	

- **SOP EXPRESSION: $M_1 + M_2 = X'Y + XY'$**
- **CANONICAL SOP FORM: $F(X, Y) = \sum M(1, 2) = X'Y + XY'$**

(II) WRITE THE POS EXPRESSION FOR B.

x	y	D	Max Terms
0	0	0	$x+y$
0	1	1	
1	0	0	$x'+y$
1	1	0	$x'+y'$

- POS EXPRESSION: $M_0 \cdot M_2 \cdot M_3 = (X+Y) \cdot (X'+Y) \cdot (X'+Y')$
- CANONICAL SOP FORM: $F(X, Y) = \prod M(0, 2, 3) = (X+Y) \cdot (X'+Y) \cdot (X'+Y')$

WRITE THE SOP EXPRESSION CORRESPONDING TO THE FOLLOWING TRUTH TABLE:

A	B	C	D (Output)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

- SOP expression: $A'B'C + A'BC + AB'C + ABC$

EXPRESS IN THE PRODUCT OF SUM (POS) FORM, THE BOOLEAN FUNCTION: F (A, B, C). THE TRUTH TABLE FOR WHICH IS GIVEN BELOW:

A	B	C	F (Output)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

- **SOP expression: $(A+B+C) \cdot (A+B'+C) \cdot (A'+B+C) \cdot (A'+B'+C)$**

CONVERT THE FOLLOWING CARDINAL FORM OF EXPRESSION INTO CANONICAL FORM:

$$F(P, Q, R) = \Pi(1, 3)$$

- **BINARY OF 1 IS 001:** $P + Q + R'$

BINARY OF 3 IS 011: $P + Q' + R'$

- **CANONICAL FORM:** $F(P, Q, R) = \Pi M(1, 3) = (P + Q + R') \cdot (P + Q' + R')$

PRINCIPAL OF DUALITY

- A Boolean expression can be converted into another form by replacing:
 - ✓ each plus(+) sign with a dot (·) and each dot sign with a plus,
 - ✓ each 1 with a 0 and each 0 with a 1.
- The expression so obtained is known as “dual of the given Boolean expression” and
- The process of conversion is termed as “Principle of Duality”.
- **Example:** Dual of $(P + Q') \cdot R \cdot 1 = P \cdot R + Q' \cdot R$ is:

$$(P \cdot Q') + R + 0 = (P + R) \cdot (Q' + R)$$

WHY DO WE SIMPLIFY BOOLEAN EXPRESSIONS?

- Reduces the number of logic gates.
- Lowers hardware cost and chip area.
- Decreases power consumption and heat generation.
- Minimizes propagation delay.
- Increases overall system speed and reliability.

MINIMIZE THE FUNCTION:

$$F(A, B, C) = A'BC' + A'BC + ABC' + ABC$$

Group the first two terms: $A'BC' + A'BC = A'B(C' + C)$ (Using Distributive Law)

Group the last two terms: $ABC' + ABC = AB(C' + C)$ (Using Distributive Law)

$$\text{So, } F = A'B(C' + C) + AB(C' + C)$$

$$\text{So, } F = A'B(1) + AB(1) \quad (\text{Using Complement Law})$$

$$\text{So, } F = B(A' + A) = B(1) \quad (\text{Using Complement Law})$$

$$\text{Hence, } F = B$$

$$F(A,B,C,D)=(A'+C)(A'+C')(A'+B+C'D)$$

► **Simplify the first two brackets:**

$$= (A' + C)(A' + C') = A'A' + A'C' + CA' + CC'$$

$$= A' + A'C' + CA' + 0 \quad (\text{using the identity \& complement laws})$$

$$= A' + A'(C' + C) \quad (\text{using the distributive law})$$

$$= A' + A'(1) \quad (\text{using the complement law})$$

$$= A' + A' = A' \quad (\text{using the idempotent law})$$

Hence, $F = A'$

$$F(A, B, C) = AB + AB' + A'BC$$

$$\diamond F = A + BC$$

$$F(A, B, C) = A'B + BC' + BC + AB'C'$$

Answer: $F = B + AC'$

$$F(A, B, C) = \Sigma m(0, 2, 4, 6, 7)$$

$$\diamond F = A'B'C' + A'BC' + AB'C' + ABC' + ABC$$

$$\diamond \text{ Minimized Form: } F = C' + ABC = C' + AB \quad (\text{Using Absorption Law})$$

Simplify the expression: $P'(P + QR) + (PC + Q'R)$.

a) $(PQ'R + QR')$

b) $(P'Q + R')$

c) $(P + QR)$

d) PC

Find the simplified term $A' (M' + A') (M + M'A)$?

a) MA'

b) $M'A$

c) $M + A$

d) $M'A'$

Minimize the Boolean expression: $F(J, K, L) = (J + KL')(JK' + L)$

a) $J + K + L'$

b) $JL' + K$

c) $K + JL$

d) $J(K' + L)$

The minimized form of Boolean logic expression

$(A'B'C' + A'BC' + A'BC + ABC')$ can be reduced to:

A: $A'C' + BC' + A'B$

B: $A'C' + B'C' + A'B$

C: $A'C + BC + A'B$

D: $AC + BC' + AB$